

UNION in DBMS

Introduction

UNION is a SQL operator used to combine the results of two or more SELECT queries into a single result table.

It removes duplicate rows automatically.

Definition

UNION combines rows from multiple queries and returns only unique records.

Syntax

```
SELECT column1, column2  
FROM table1
```

UNION

```
SELECT column1, column2  
FROM table2;
```

Rules of UNION

1. Number of Columns Must Be Same

Both SELECT statements must contain the same number of columns.

Example

```
SELECT id, name FROM students  
UNION  
SELECT id, name FROM teachers;
```

2. Data Types Must Be Compatible

The corresponding columns should have similar data types.

Correct Example

```
SELECT id FROM students  
UNION  
SELECT teacher_id FROM teachers;
```

3. Duplicate Rows Are Removed

UNION automatically removes duplicate records.

Example of UNION

Table: Students_A

Name

Ravi

Sita

Table: Students_B

Name

Ravi

Kumar

Query

```
SELECT name FROM Students_A  
UNION  
SELECT name FROM Students_B;
```

Output

Name

Ravi

Sita

Kumar

Duplicate value "Ravi" appears only once.

UNION ALL

UNION ALL combines records from multiple queries but keeps duplicate values.

Syntax of UNION ALL

```
SELECT column_name FROM table1  
UNION ALL  
SELECT column_name FROM table2;
```

Example of UNION ALL

Query

```
SELECT name FROM Students_A  
UNION ALL  
SELECT name FROM Students_B;
```

Output

Name

Ravi

Sita

Ravi

Kumar

Duplicate values are not removed.

Difference Between UNION and UNION ALL

UNION

Removes duplicates

Slower

Returns unique records

UNION ALL

Keeps duplicates

Faster

Returns all records

Example 1: Student Lists

Table: ClassA

id name

1 Ravi

2 Sita

Table: ClassB

id name

3 Kumar

4 Priya

Query

```
SELECT name FROM ClassA  
UNION  
SELECT name FROM ClassB;
```

Output

name

Ravi

Sita

Kumar

Priya

Example 2: Employee Records

```
SELECT employee_name FROM company1  
UNION  
SELECT employee_name FROM company2;
```

This query combines employee names from two companies.

ORDER BY with UNION

ORDER BY is used at the end of the UNION query.

Example

```
SELECT name FROM students  
UNION  
SELECT name FROM teachers  
ORDER BY name;
```

This sorts the final combined result.

Advantages of UNION

1. Combines Multiple Results

UNION merges data from multiple tables or queries.

2. Removes Duplicate Records

Duplicate rows are automatically removed.

3. Easy Data Analysis

Helps analyze data from different sources together.

Disadvantages of UNION

1. Same Columns Required

Both queries must have the same number of columns.

2. Slower Than UNION ALL

UNION takes more time because it checks for duplicates.

Real-Time Examples

Scenario	Usage
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Student records	Combine students from different classes
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Employee lists	Combine employees from branches
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Product records	Combine products from stores
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Common Errors in UNION

Error 1: Different Number of Columns

Wrong Query

```
SELECT id, name FROM students
UNION
SELECT id FROM teachers;
```

Reason

The number of columns is different.

Error 2: Incompatible Data Types

Wrong Query

```
SELECT name FROM students
UNION
SELECT salary FROM employees;
```

Reason

String and integer data types are not compatible.

Interview Questions

1. What is UNION in DBMS?

UNION combines the results of multiple SELECT queries into one result.

2. What is the difference between UNION and UNION ALL?

UNION removes duplicates, while UNION ALL keeps duplicates.

3. Can UNION combine different tables?

Yes, if the number of columns and data types are compatible.

shyam-sudheer1602